

RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

128 MiB Octal NOR

File: nor_flash.kicad_sch

RABP1 IO allocation

SDRAM_CS_N
SDRAM_RAS_N
SDRAM_WE_N
SDRAM_CKE
SDRAM_CAS_N
SDRAM_A[0..12]
SDRAM_DQ[0..31]
SDRAM_CLK
SDRAM_DQM[0..3]
SDRAM_BA[0..1]

File: mcu_interfaces.kicad_sch

RABP1 internal core regulator

File: mcu_core_power.kicad_sch

RABP1 IO and analog supplies

File: mcu_supply_power.kicad_sch

RABP1 clocks, reset and debug

File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

Page and volume buttons

PAGE_PREV_N
PAGE_NEXT_N
VOL_DOWN_N
VOL_UP_N

File: user_controls.kicad_sch

ESP32-C6 radio

RADIO_CS_N
RADIO_COPI
RADIO_SCLK
RADIO_CIPO
RADIO_RESET_REQ_N
RADIO_PWR_EN
MCU_RESET_N

File: radio_esp32.kicad_sch

64 MiB SDRAM

SDRAM_DQM[0..3]
SDRAM_A[0..12]
SDRAM_WE_N
SDRAM_DQ[0..31]
SDRAM_CS_N
SDRAM_CKE
SDRAM_CAS_N
SDRAM_RAS_N
SDRAM_BA[0..1]
SDRAM_CLK

File: sdram.kicad_sch

Power button and source control

POWER_BUTTON_N
POWER_KILL_N

File: power_button.kicad_sch

Main 3V3 digital supply

File: main_3v3_supply.kicad_sch

MAIN_PWR_EN

Sheet: /
File: ereader_rev1.kicad_sch

Title: RA8P1 e-reader – schematic index

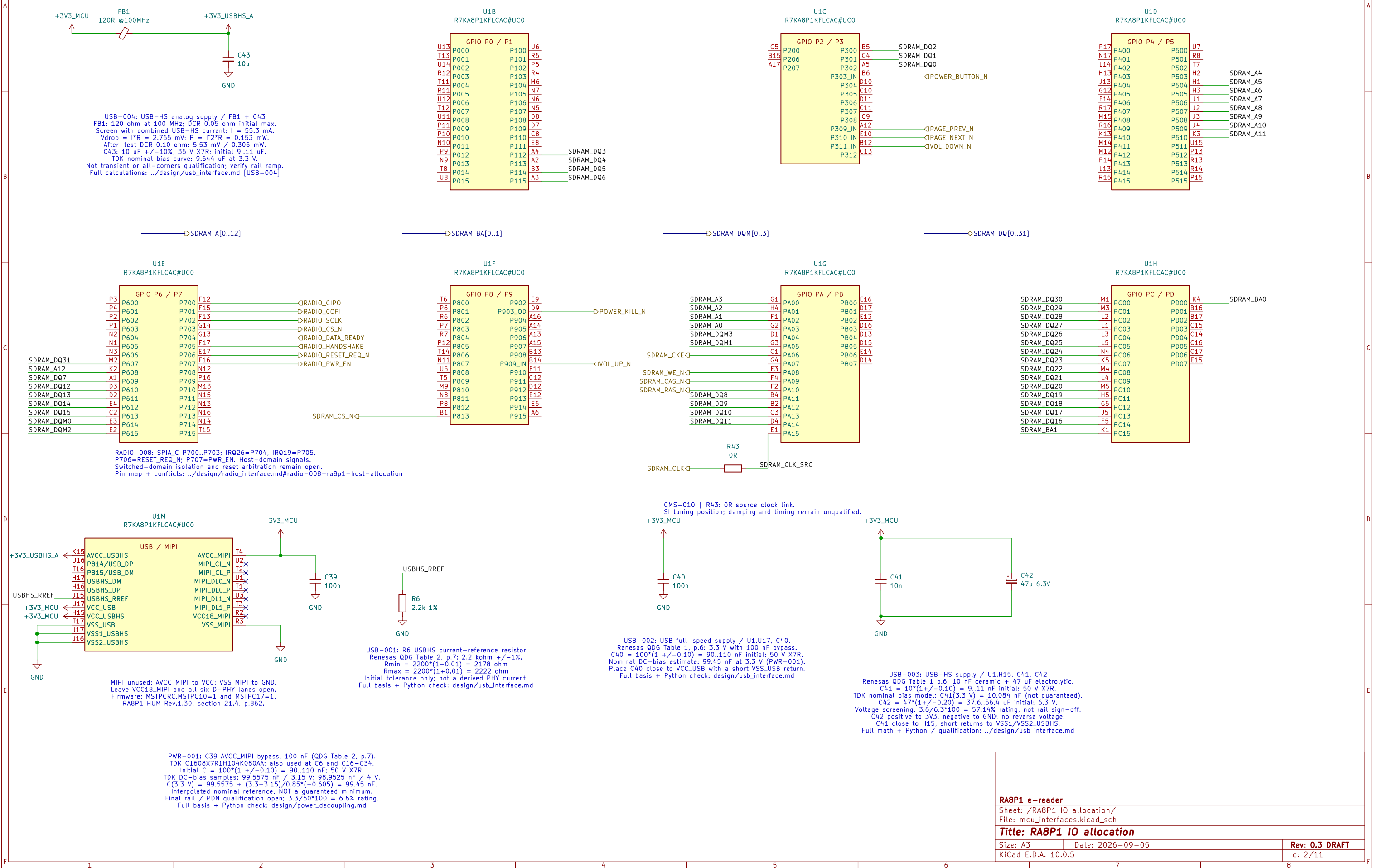
Size: A3 Date: 2026-09-05

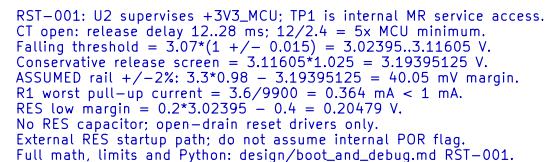
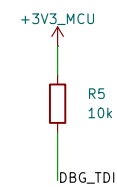
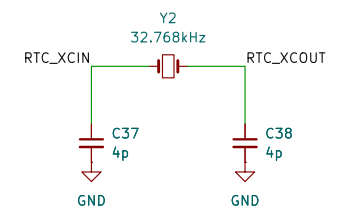
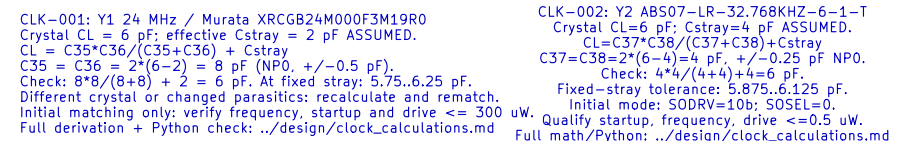
KiCad E.D.A. 10.0.5

Rev: 0.3 DRAFT

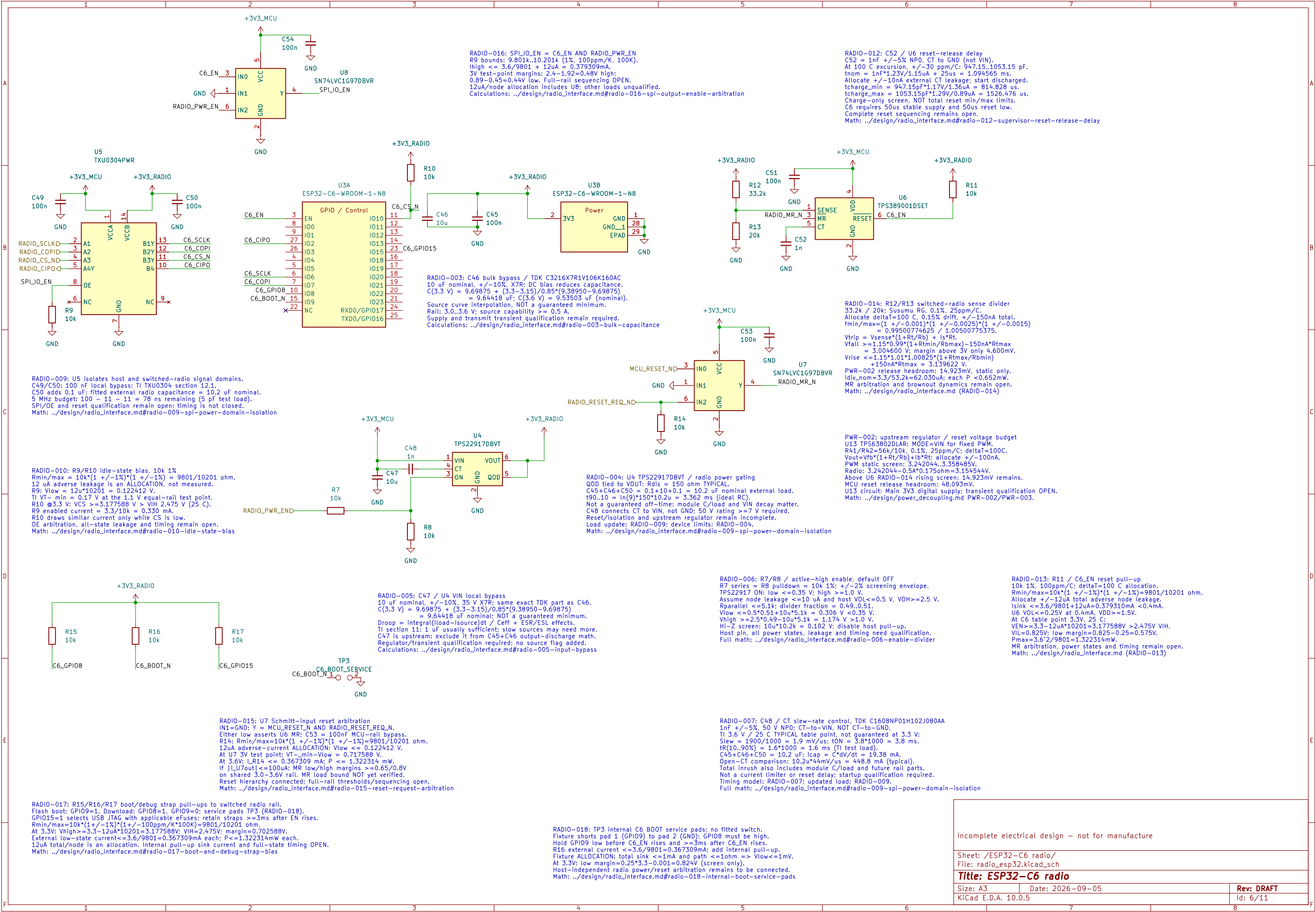
Id: 1/11

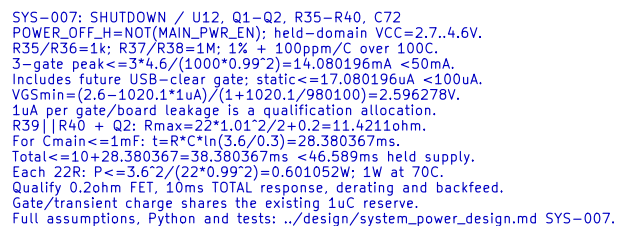
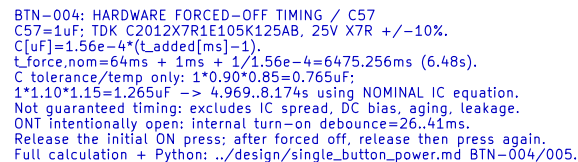
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



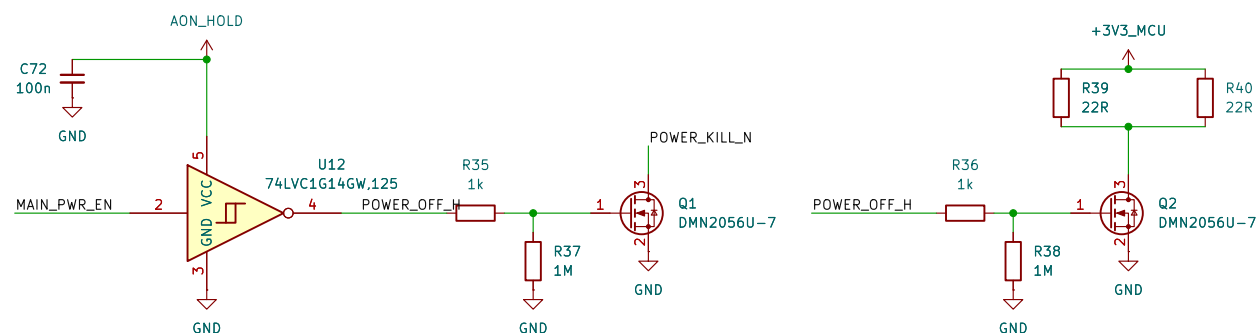
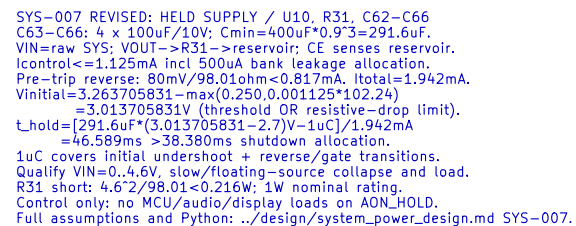


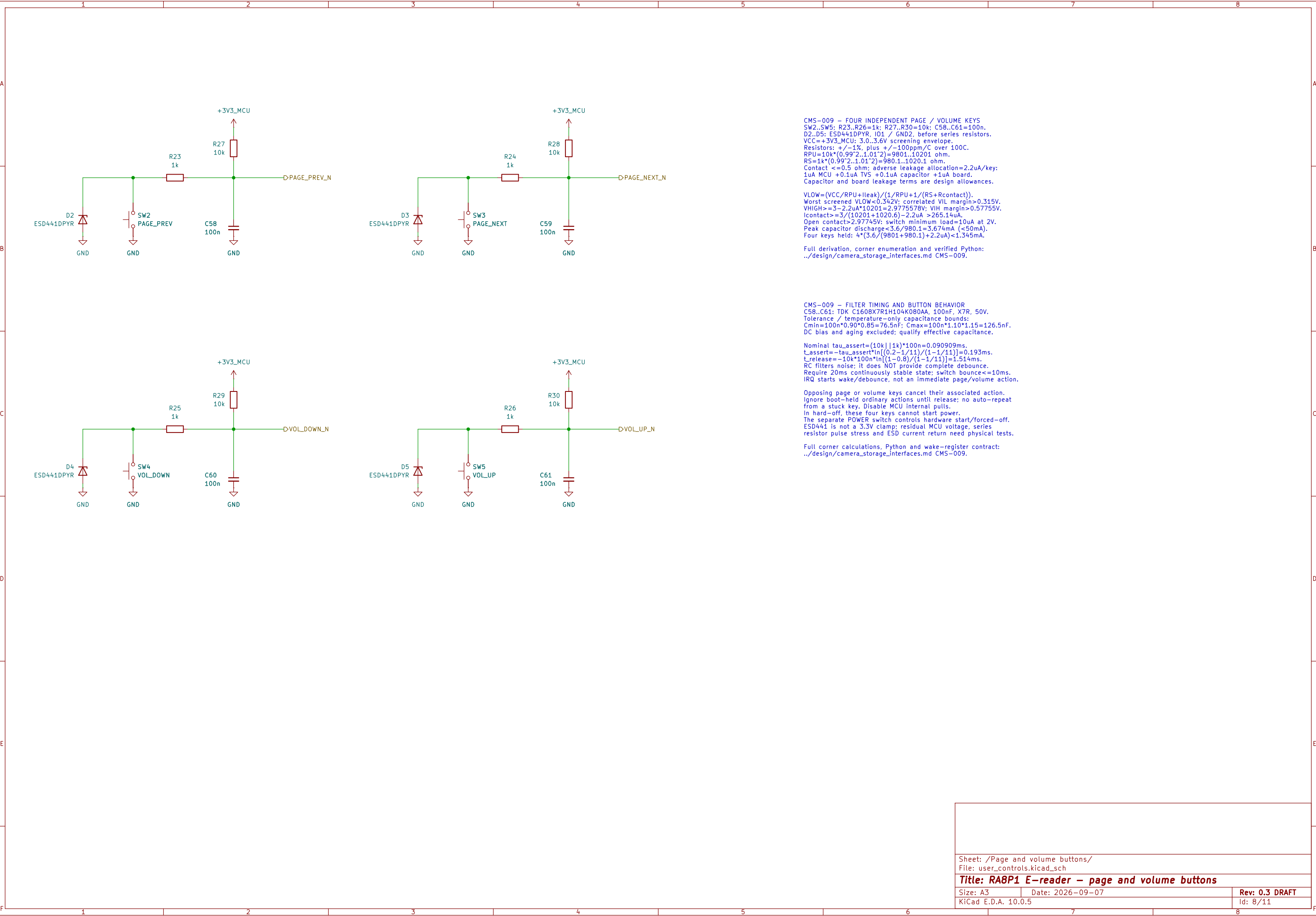
References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3–4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.





ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$V_{LOW} = (VCC/RPU + I_{leak}) / (1/RPU + 1/(RS + R_{contact}))$.
Worst screened $V_{LOW} < 0.342V$; correlated VIL margin > 0.315V.
 $V_{HIGH} > 3 - 2.2uA * 10201 = 2.9775578V$; VIH margin > 0.57755V.
 $I_{contact} > 3 / (10201 + 1020.6) - 2.2uA > 265.14uA$.
Open contact > 2.97745V; switch minimum load = 10uA at 2V.
Peak capacitor discharge < 3.6/980.1 = 3.674mA (< 50mA).
Four keys held: $4 * (3.6 / (9801 + 980.1) + 2.2uA) < 1.345mA$.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
 $C_{min} = 100n * 0.90 * 0.85 = 76.5nF$; $C_{max} = 100n * 1.10 * 1.15 = 126.5nF$.
DC bias and aging excluded; qualify effective capacitance.

Nominal $\tau_{assert} = (10k || 1k) * 100n = 0.090909ms$.
 $\tau_{assert} = -\tau_{assert} * \ln[(0.2 - 1/11) / (1 - 1/11)] = 0.193ms$.
 $\tau_{release} = -10k * 100n * \ln[(1 - 0.8) / (1 - 1/11)] = 1.514ms$.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce <= 10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

Sheet: /Page and volume buttons/
File: user_controls.kicad_sch

Title: **R8P1 E-reader – page and volume buttons**

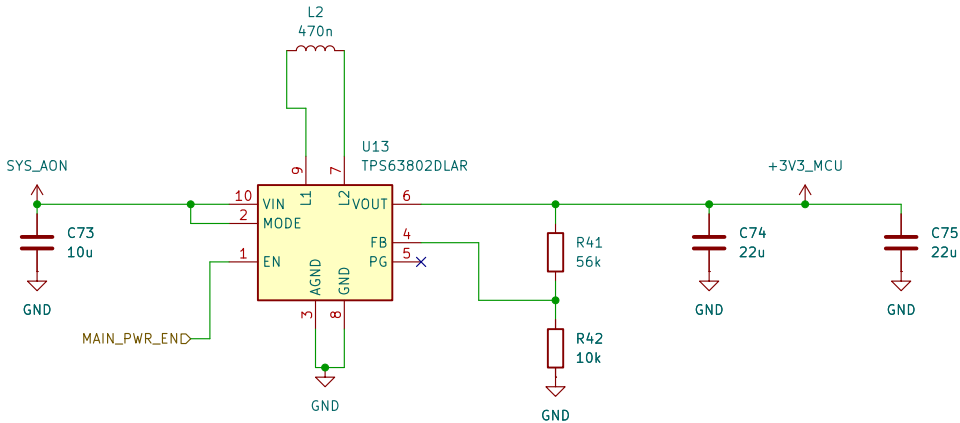
Size: A3 Date: 2026-09-07

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

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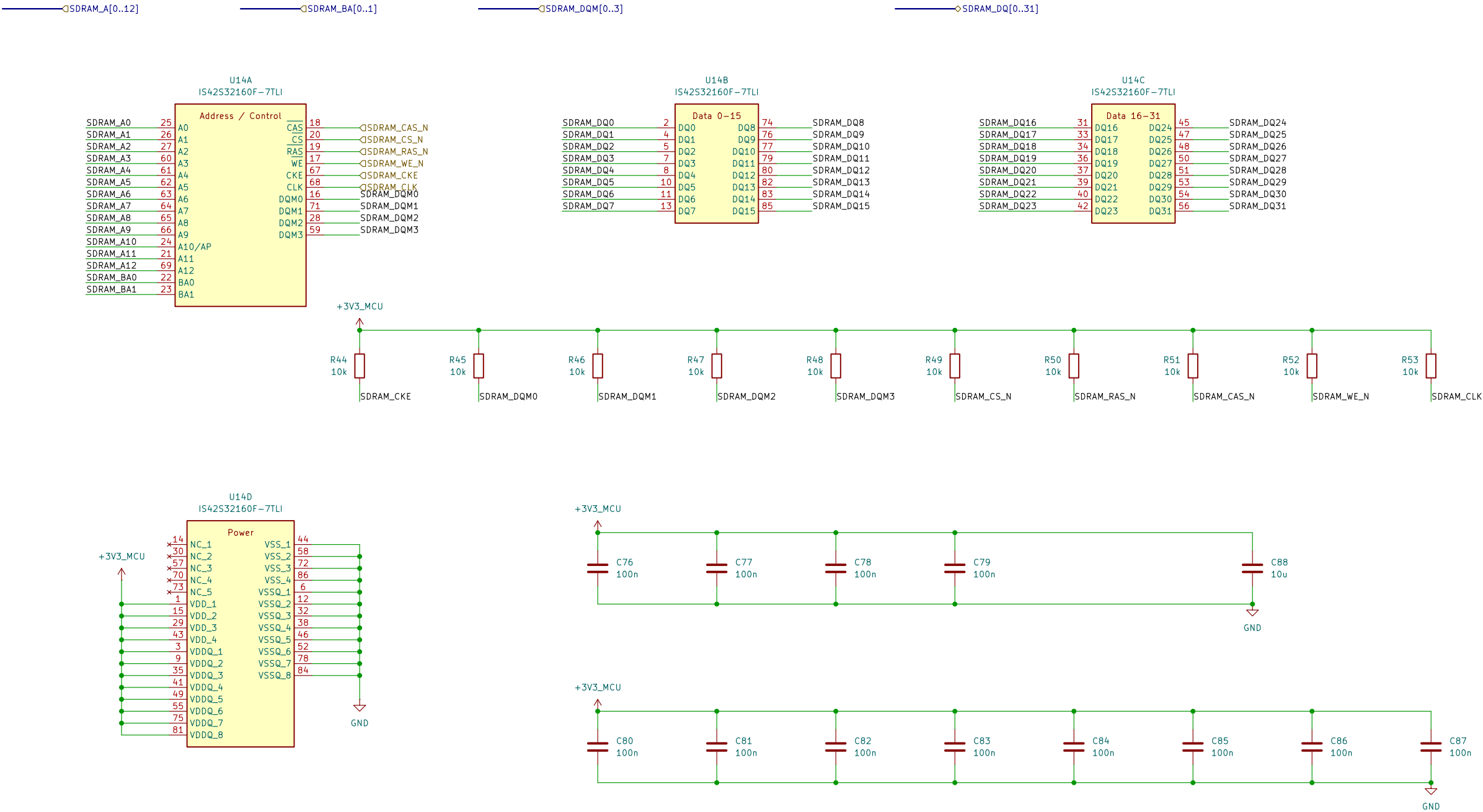
MAIN DIGITAL 3.3 V SUPPLY
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS
U13 TPS63802DLAR; MODE=VIN, fixed PWM.
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.
Vnom=0.500*(1+56k/10k)=3.300V.
Idivider=3.300/(56k+10k)=50uA.
For 100C resistor–temperature change:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vmin=0.495*(1+56k*fLo/(10k*fHi))–100nA*56k*fHi
=3.242044V.
Vmax=0.505*(1+56k*fHi/(10k*fLo))+100nA*56k*fHi
=3.358485V.
Static screen only; ripple, load steps and drift still require validation.
C73: CIN_model=10u*0.90*0.85*0.60=4.590uF >=4uF.
C74+C75: COUT_model=2*22u*0.90*0.85*0.60=20.196uF >=7uF.
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS
Main continuous allocation: 0.750 MCU + 0.500 radio
+ 0.250 SDRAM + 0.250 NOR + 0.050 logic = 1.800A.
These are qualification allocations, not measured maximum currents.
SD, camera, audio, display/touch/frontlight need separate SYS supplies.
Pout=3.358485094*1.80=6.045273W.
VIN>=3.20V allocation; raw trip min=3.263705831V.
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.
At ASSUMED eta=85%: Iin=Pout/(eta*3.20)=2.222527A.
Ploss=Pout*(1/eta–1)=1.066813W.
Reference rise=Ploss*81C/W=86.411846C; NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.550 other = 1.880A.
Radio on would give 2.380A > nominal 2A; keep radio OFF through start/wake.
Largest wake–reference subtotal=1.270+0.550=1.820A.
DCDC figures are reference–only; non–DCDC start and extra charging excluded.
Additional charging: I=Cadditional*dV/dt; qualify the actual ramp.
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.
No externally driven +3V3_MCU: forced PWM can sink reverse current.
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: ../design/power_decoupling.md PWR–003;
source–loss shutdown: ../design/system_power_design.md SYS–007.

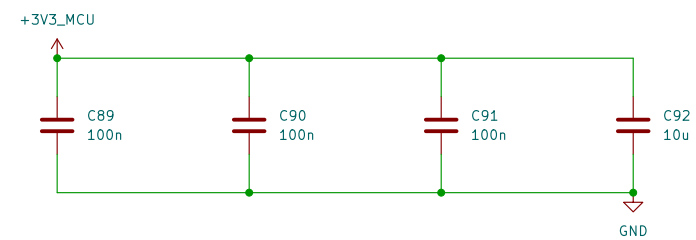
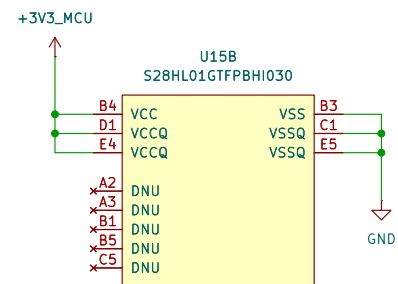
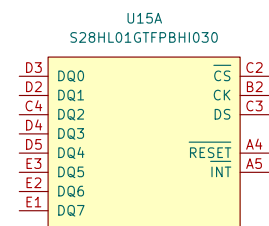
64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail;
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.



CMS-011 | NOR supply and bypass.
 U15B VCC and VCCQ share +3V3_MCU (2.7 to 3.6 V).
 C89-C91: 3 x 100 nF, one per supply ball.
 C92: 10 uF local bulk.
 Cnom = $3 \times 0.1 + 10 = 10.3 \mu\text{F}$.
 Initial $\pm 10\%$: $C_{\text{min}}/\text{max} = 10.3 \times (0.9 / 1.1)$
 $= 9.27 / 11.33 \mu\text{F}$ - not an effective-C guarantee.
 Included in the PWR-003 1 mF whole-rail limit.
 DC bias, temperature, aging and PDN qualification OPEN.
 Linked CMS-011B/F: sources and Python-verified arithmetic.